



Course Code & Title	ETCCCM102 (Computational Mathematics II)
Programme & School	B.Tech CSE (School of Engineering and Technology)
Faculty Name	Dr. Vishal
Type of Assignment	Handwritten and Upload
Weightage / Marks	10
Submission Mode	LMS

Assignment 2: Numerical Methods

Instructions:

- Solve all questions showing detailed steps.
- Use A4 size sheets.
- Submit neatly written solutions.

Questions:

1. Show that the equation

$$f(x) = x^3 + 4x^2 - 10 = 0$$

has a root in the interval $[1, 2]$.

[CO1]

2. Solve the equation

$$x^3 - 1.1x^2 + 4x - 4.4 = 0$$

correct to two significant figures using the **Bisection Method**.

[CO1]

3. Use the **Secant Method** to solve

$$xe^{0.5x} + 11.2x = 10x + 5$$

using $x_i = 1.6$ and $x_{i-1} = 10$. Perform calculations up to three iterations. [CO1]

4. Use the **Secant Method** (two iterations) to find a root of

$$f(x) = 3e^{-x} - 3x = 0$$

in the interval $[0, 1]$. Take $x_0 = 0$, $x_1 = 1$.

[CO1]

5. Using **Newton's Method**, determine x_2 for

$$f(x) = x^3 - 7x^2 + 8x - 3$$

given that $x_0 = 5$. [CO2]

6. Using **Newton's Method**, find the root of

$$x^4 - 5x^3 + 9x + 3 = 0$$

that lies in the interval $[4, 6]$, correct to six decimal places. [CO2]

7. Using **Newton's Method**, solve

$$2x^2 + 5 = e^x$$

for a root in the interval $[3, 4]$, correct to six decimal places. [CO2]

8. Determine the root of

$$f(x) = 2x^3 - 11.7x^2 + 17.7x - 5$$

using the **Fixed Point Iteration Method** with $x_0 = 3$. Perform three iterations. [CO2]

9. Determine a root of

$$x^3 - x - 1 = 0$$

in the interval $[1, 2]$ using **Fixed Point Iteration**, given $x_0 = 1.5$. Perform four iterations. [CO2]

10. Find the **LU decomposition** of the matrix

$$\begin{bmatrix} 3 & 1 \\ -6 & -4 \end{bmatrix}.$$

[CO4]

11. Find the **LU decomposition** of the matrix

$$\begin{bmatrix} 3 & 1 & 6 \\ -6 & 0 & -16 \\ 0 & 8 & -17 \end{bmatrix}.$$

[CO4]

12. Use the **Cholesky Decomposition Method** to solve the system of equations

$$\begin{aligned}4x_1 + 2x_2 + 14x_3 &= 14, \\2x_1 + 17x_2 - 5x_3 &= -101, \\14x_1 - 5x_2 + 83x_3 &= 155.\end{aligned}$$

[CO4]

13. Solve the following system using the **Gauss–Seidel Method**

$$\begin{aligned}45x_1 + 2x_2 + 3x_3 &= 58, \\-3x_1 + 22x_2 + 2x_3 &= 47, \\5x_1 + x_2 + 20x_3 &= 67.\end{aligned}$$

[CO4]

14. Using the **Gauss–Seidel Method**, solve

$$\begin{aligned}27x + 6y - z &= 85, \\6x + 15y + 2z &= 72, \\x + y + 54z &= 110.\end{aligned}$$

[CO4]

15. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}.$$

Find the **Singular Value Decomposition** of A , that is, determine matrices U , Σ , and V such that $A = U\Sigma V^T$. [CO4]

16. Compute the **Singular Value Decomposition** of the 2×3 matrix

$$A = \begin{bmatrix} 2 & 1 & 4 \\ -1 & 2 & -2 \end{bmatrix}.$$

[CO4]

17. Solve the following system using the **Jacobi Iteration Method**

$$\begin{aligned}20x + y - 2z &= 17, \\3x + 20y - z &= -18, \\2x - 3y + 20z &= 25.\end{aligned}$$

[CO4]

18. Use the **Gauss–Jacobi Method** to solve the following system of linear equations. Show the values of x_1, x_2, x_3 , and x_4 for the first four iterations. Assume the initial approximation $x_1 = x_2 = x_3 = x_4 = 0$.

$$12x_1 + 3x_2 - 4x_3 + x_4 = 10,$$

$$4x_1 + 9x_2 + 2x_3 - 2x_4 = 20,$$

$$2x_1 - 3x_2 + 10x_3 + x_4 = 30,$$

$$-3x_1 + 2x_2 - 3x_3 + 12x_4 = 40.$$

[CO4]